

Phishing Email Detection & Awareness System

1. Executive Summary

Phishing is one of the most common and financially damaging cyber threats affecting businesses, students, government institutions, and individual users worldwide. Attackers often impersonate trusted brands, payment services, banks, or internal departments to trick victims into clicking malicious links, revealing passwords, sharing one-time passwords (OTPs), or downloading malware.

The purpose of this report is to investigate phishing emails, identify common warning signs, classify the threat level, and provide practical awareness guidance that organizations can use to improve employee security behavior.

Three phishing case studies were analyzed in this report, including impersonation attempts involving Instagram, PayPal, and Wells Fargo. Each case demonstrates how attackers use urgency, fake sender domains, brand logos, emotional pressure, and deceptive links.

This report concludes that phishing prevention depends strongly on awareness, verification habits, technical email filtering, and multi-factor authentication.

2. Introduction

Cybersecurity threats continue to evolve rapidly, but phishing remains effective because it targets human trust rather than system vulnerabilities. A phishing attack usually begins with a fake email or message that appears legitimate. The attacker's objective is to convince the victim to take immediate action.

Modern phishing emails are often professionally designed. They may contain company logos, realistic formatting, legal disclaimers, and convincing language. Because of this, many users click malicious links before carefully checking the sender or destination.

Organizations face major risks from phishing:

- Credential theft

- Financial fraud
- Business email compromise
- Data leakage
- Malware infection
- Reputation damage
- Operational downtime

Therefore, phishing awareness training is a critical component of security programs.

3. Objectives

The objectives of this task are:

- Analyze phishing email samples from real-world style scenarios
 - Detect suspicious domains and malicious intent
 - Identify social engineering techniques
 - Classify risk levels clearly
 - Explain threats in simple language
 - Create awareness guidance for employees
 - Recommend controls for organizations
 - Improve practical cybersecurity knowledge
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4. Understanding Phishing Attacks

Phishing is a cybercrime method where attackers impersonate trusted entities to steal sensitive data or gain unauthorized access.

Typical Attack Flow

1. Attacker creates fake email
2. Uses trusted brand identity
3. Sends urgency message
4. Victim clicks link
5. Victim enters credentials on fake page
6. Credentials stolen
7. Attacker accesses real account

Why Phishing Works

- Users act quickly under pressure
 - Emails appear official
 - Users trust logos and branding
 - Many users do not inspect URLs
 - Fear-based messages reduce judgment
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5. Common Types of Phishing

5.1 Email Phishing

Mass emails sent to many victims pretending to be trusted companies.

5.2 Spear Phishing

Targeted phishing using personal or company-specific details.

5.3 Whaling

Phishing targeting executives or senior leadership.

5.4 Smishing

SMS text messages containing malicious links.

5.5 Vishing

Voice phishing using fake calls from banks or support teams.

5.6 Clone Phishing

A legitimate email is copied and modified with malicious links.

6. Email Sample Investigation

Case Study 1: Fake Instagram Security Alert

Subject:

Unusual Login Detected – Verify Now

Observations:

- Sender used free email domain (gmail.com)
- Pretended to be Instagram security
- Claimed unusual login attempt
- Requested immediate verification
- Included suspicious external URL

Indicators:

- Fake sender identity
- Urgency tactic
- Fear of account compromise
- Credential theft attempt

Risk Level:

High Risk Phishing

Recommended Action:

Delete email, report phishing, verify account only through official app.

Proof:

Unusual Login Detected - Verify Now Inbox x



instasecurityy@gmail.com

to me ▼

Dear satvik____17,

We've detected unusual login activity on your **Instagram** account from a new device/location.

To prevent unauthorized access, please verify your identity immediately:



[Secure **Instagram** Login]: <https://instaa-d60da.web.app>

If you didn't attempt to log in recently, someone may be trying to access your account.

This link expires in 24 hours.

Thank you,

Instagram Security Team

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Case Study 2: Fake PayPal Account Limited Notice

Subject:

[PayPal]: Your account access has been limited

Observations:

- Uses PayPal logo
- Generic greeting (“Dear PayPal customer”)
- Threatens account disablement in 24 hours
- Requests confirmation of information

Indicators:

- Brand impersonation
- Panic pressure tactic
- Generic greeting
- Possible fake landing page

Risk Level:

High Risk Phishing

Proof:

[PayPal]: Your account access has been limited

Team Support services@paypal-accounts.com
to me



Dear PayPal customer,

Your PayPal account is limited, You have 24 hours to solve the problem or your account will be permanetly disabled.

We are sorry to inform you that you no longer have access to PayPal's advantages like purchasing, and sending and receiving money.

Why is my PayPal account limited?

We believe that your account is in danger from unauthorized users.

What can I do to resolve the problem?

You have to confirm all of your account details on our secured server by clicking the link below and following the steps.

[Confirm Your Information](#)

Case Study 3: Fake Wells Fargo Security Alert

Subject:

!Alerts!

Observations:

- Fake sender address
- Claims account locked
- Requests clicking unlock link
- Uses bank branding

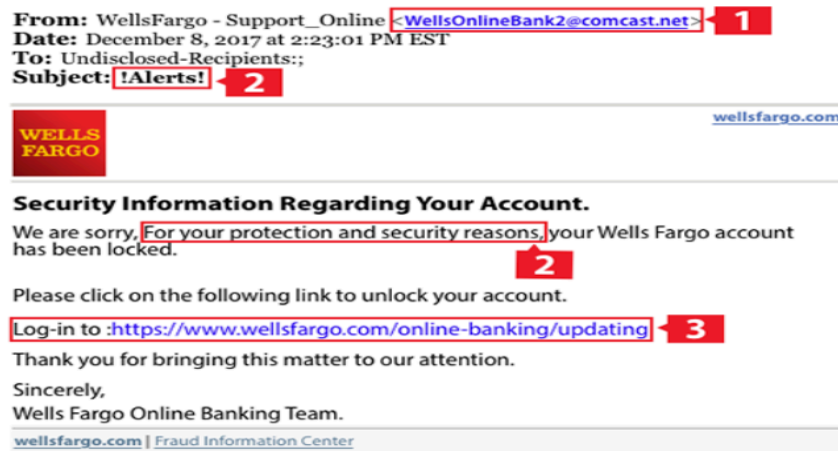
Indicators:

- Financial institution spoofing
- Dangerous credential theft attempt
- Suspicious urgency

Risk Level:

High Risk Banking Phishing

Proof:



7. Technical Indicators of Phishing

Indicator	Description	Risk
Fake Domain	Slight misspelling or unrelated sender	High
Generic Greeting	Dear Customer / Dear User	Medium
Urgency	Act now / 24 hours left	High
Threat Language	Locked account / disabled account	High
Suspicious URL	Different destination domain	High
Grammar Errors	Poor writing quality	Medium
Unexpected Attachment	ZIP / EXE / macro files	Critical

Domain Examples

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8. Risk Classification Model

A simple scoring model was designed for classification.

Factor	Score
Fake sender	+2
Suspicious link	+3
Urgency language	+2
Generic greeting	+1
Credential request	+3
Attachment	+2

Final Score Result

Score	Classification
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0–2	Safe
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3–5	Suspicious
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6+	Phishing
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All three case studies scored above 6.

9. Prevention & Awareness Guidelines

For Individual Users

- Verify sender domain carefully
- Hover over links before clicking
- Use official apps/sites directly
- Never share OTP or passwords
- Use unique passwords
- Enable multi-factor authentication
- Keep browser and device updated

Safe Behavior Example

Instead of clicking a bank email link, manually type the bank website.

Warning Behavior Example

Clicking urgent messages without checking sender details.

10. Organizational Recommendations

Organizations should implement layered phishing defense strategies.

Awareness Controls

- Monthly phishing awareness sessions
- Simulated phishing campaigns
- Security newsletters
- New employee onboarding training

Technical Controls

- Email spam filtering
- DMARC / SPF / DKIM protections
- Web filtering
- Endpoint protection
- Multi-factor authentication

Incident Response Controls

- One-click report phishing button
 - Security escalation process
 - Account reset workflow
 - Threat intelligence review
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11. Conclusion

Phishing remains successful because it exploits human emotions such as fear, urgency, curiosity, and trust. Even technically secure environments may suffer breaches if users are not trained to identify deceptive emails.

The three investigated samples demonstrate common phishing patterns: fake domains, impersonation of trusted brands, urgency messaging, and credential collection attempts.

The best defense combines:

- User awareness
- Verification habits
- Technical filtering

- Strong authentication
- Fast reporting culture

Organizations that regularly educate users and improve email security controls significantly reduce phishing risk.

12. References

- Google Message Header Analyzer
- MXToolbox
- VirusTotal
- Public phishing awareness materials
- Security best practice guidance